

FREE LEAD MAGNET · BIOAI LAB · DEEPIOTICS

The Bioinformatics Interview Bible

20 Real Interview Questions · 20 Model Answers · Zero Blanks
Written in simple English every biology student can understand

20

questions & answers

Interview Prep

Bioinformatics

BioCode Mastery

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You Already Know More Than You Think.

Most students walk into bioinformatics interviews feeling nervous. They have studied hard. They understand the concepts. But the moment an interviewer asks "explain how GATK HaplotypeCaller works," the brain goes blank.

This guide will fix that. It has 20 real questions that come up in bioinformatics interviews. Each one has a clear, simple answer — written the way a confident analyst would say it, not the way a textbook writes it. Read each answer. Understand it. Say it aloud. Then you will own it.

The questions cover 8 topics: NGS basics, quality control, alignment, variant calling, RNA-Seq, databases, AI in biology, and clinical applications. These are the same topics that appear in interviews at research labs, biotech companies, and hospitals around the world.

Who this guide is for

You if you have an interview coming up and you want to feel ready. **You** if you studied bioinformatics but the interview questions feel different from what you learned. **You** if you want to say the right things without sounding like you are reading from a textbook. **You** if you are applying for any role that involves NGS data, genomics, or biological data analysis.

This guide has helped students get interview calls at top research labs and biotech companies.

■ Read each question first — then read the answer

■ Say the answer aloud — hearing yourself helps

■ Read the TIP box — it tells you what impresses

■ Add your own examples to make it personal

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NGS FUNDAMENTALS

**What is NGS and why is it better than old sequencing?**

NGS stands for Next Generation Sequencing. It can read millions of pieces of DNA at the same time. Old sequencing (Sanger) could only read one piece at a time. NGS is much faster and cheaper. That is why labs use it for things like finding mutations, studying gene activity, and whole genome analysis.

INTERVIEW TIP: Read one piece at a time = OLD. Read millions at once = NGS. That is the key difference.

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NGS FUNDAMENTALS

**What are the main steps in an NGS experiment?**

There are 6 steps. First, you collect DNA or RNA from a sample. Second, the machine reads the DNA and creates raw data. Third, you check if the data quality is good. Fourth, you match the data to a reference genome. Fifth, you find the differences between your sample and the reference. Sixth, you figure out what those differences mean for health or biology.

INTERVIEW TIP: The 6 steps: Collect → Read → Check quality → Match to reference → Find differences → Understand meaning.

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NGS FUNDAMENTALS

**What is a FASTQ file? What is inside it?**

A FASTQ file is the raw output from an NGS machine. Every DNA piece (called a read) takes up 4 lines. Line 1 starts with @ and gives the read a name. Line 2 is the actual DNA letters (A, T, G, C). Line 3 is just a + sign. Line 4 gives a quality score for each DNA letter — it tells you how confident the machine was when reading each base.

INTERVIEW TIP: 4 lines per read: Name → DNA letters → + sign → Quality scores. Remember this order.

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NGS FUNDAMENTALS

**What is a Phred quality score? What number is good?**

A Phred score (Q score) tells you how accurate each DNA base reading is. Q20 means the machine has a 1 in 100 chance of making an error. Q30 means only a 1 in 1000 chance of error. Q30 is the gold standard. If your bases have Q scores below 20, they should be removed because they are not reliable enough for analysis.

INTERVIEW TIP: Q30 = very good. Q20 = okay. Below Q20 = not reliable. Always aim for Q30 in your data.



What is the difference between DNA sequencing and RNA sequencing?

DNA sequencing reads the permanent genetic code of a cell — it does not change. RNA sequencing (RNA-Seq) reads which genes are active at a specific moment in time. DNA-Seq answers: what mutations does this person have? RNA-Seq answers: which genes are switched on or off in this tissue? Both use NGS machines but they answer different questions.

INTERVIEW TIP: DNA-Seq = permanent mutations. RNA-Seq = which genes are active right now. Two different questions.

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QUALITY CONTROL & TOOLS

**What does FastQC do and what do you look for in the report?**

FastQC is a free tool that checks the quality of your raw NGS data. It creates a report with traffic light colours — green means good, orange means warning, red means problem. The most important things to check are: how clean the base quality is across the whole read, whether there is any adapter contamination, and whether some sequences appear too many times (duplication). Always run FastQC before doing any other analysis.

INTERVIEW TIP: FastQC = quality check. Green = good. Orange = check it. Red = fix it. Always run this first.

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QUALITY CONTROL & TOOLS

**What is Trimmomatic and when do you use it?**

Trimmomatic is a tool that cleans up your raw reads. You use it after FastQC shows problems. It removes adapter sequences that got attached by accident during the experiment. It also cuts off the ends of reads where the quality drops below Q20. After trimming, your data is cleaner and your analysis will be more accurate. Always run FastQC again after trimming to confirm the problems are fixed.

INTERVIEW TIP: FastQC shows a problem → run Trimmomatic → run FastQC again to check it is fixed.

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QUALITY CONTROL & TOOLS



What is a BAM file and how is it different from a SAM file?

A SAM file (Sequence Alignment Map) stores information about where each read mapped to the genome. It is a text file you can open and read. A BAM file has exactly the same information but it is compressed — it takes up much less space on your computer. In practice, always work with BAM files because whole genome SAM files can be over 100 GB. You need Samtools to work with BAM files.

INTERVIEW TIP: SAM = readable text file. BAM = same data but compressed (much smaller).
Always use BAM in practice.

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QUALITY CONTROL & TOOLS



Why do you sort and index a BAM file?

After alignment, the reads in a BAM file are in random order. Sorting puts them in order by chromosome and position. Indexing creates a small map file (BAM.bai) that lets tools jump quickly to any position in the genome without reading the whole file. Both steps are required before you can do variant calling or visualise the data. Samtools sort and samtools index are the commands used.

INTERVIEW TIP: Sort = put reads in order. Index = create a map for fast access. Both steps are needed before next steps.

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ALIGNMENT

**Why do we use BWA-MEM for DNA alignment?**

BWA-MEM is the standard tool for aligning short DNA reads to a reference genome. It is fast, accurate, and handles paired-end reads well. It is also good at finding reads that span breakpoints (split reads), which matters for detecting structural variants. For RNA-Seq data, we use HISAT2 instead because RNA reads can span the joins between exons — BWA-MEM does not understand that, but HISAT2 does.

INTERVIEW TIP: DNA alignment = BWA-MEM. RNA-Seq alignment = HISAT2. Interviewers always test if you know this difference.

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ALIGNMENT

**What is a CIGAR string in a BAM file?**

A CIGAR string describes how a read aligns to the reference genome. It uses letters to explain what happened at each position. M means the base matched (or mismatched) the reference. I means your read has extra bases not in the reference (insertion). D means your read is missing bases that are in the reference (deletion). S means some bases at the ends were not aligned (soft clip). For example, 75M means all 75 bases aligned to the reference.

INTERVIEW TIP: CIGAR letters: M = match, I = insertion in read, D = deletion in read, S = bases not aligned. Simple.

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VARIANT CALLING

**What is the difference between an SNP and an Indel?**

An SNP (Single Nucleotide Polymorphism) is when one DNA letter is different from the reference. For example, the reference has A but your sample has G at that position. An Indel is when one or more DNA letters are either added (insertion) or removed (deletion) compared to the reference. Both types of variants are important in medicine — SNPs can cause disease, and Indels can disrupt how a gene works completely.

INTERVIEW TIP: SNP = one letter changed. Indel = letters added or removed. Both can cause disease.

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VARIANT CALLING

**How does GATK HaplotypeCaller find variants?**

GATK HaplotypeCaller is the most trusted variant calling tool. It works in a smart way — instead of just looking at one position at a time, it looks at small regions of the genome and rebuilds the likely DNA sequences in that region. This makes it better at finding difficult variants and Indels. The output is a VCF file listing all the positions where your sample is different from the reference.

INTERVIEW TIP: HaplotypeCaller rebuilds local regions to find variants. Output = VCF file. This is the industry standard.



What is a VCF file? What do the main columns mean?

VCF stands for Variant Call Format. It is the standard output of variant calling. Each row is one variant. The key columns are: CHROM = which chromosome the variant is on. POS = the exact position. REF = what the reference genome has at that position. ALT = what your sample has instead. FILTER = PASS means the variant is high quality. The SAMPLE column tells you the genotype — 0/0 means same as reference, 0/1 means one copy changed (heterozygous), 1/1 means both copies changed (homozygous).

INTERVIEW TIP: VCF columns to know: CHROM, POS, REF, ALT, FILTER, SAMPLE. Genotype 0/1 = heterozygous. Always comes up.

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RNA-SEQ ANALYSIS

**What is RNA-Seq used for?**

RNA-Seq is used to measure gene expression — which genes are active and how active they are. If you compare a cancer tissue and a normal tissue, RNA-Seq can tell you which genes are more active in cancer (upregulated) and which are less active (downregulated). These are called differentially expressed genes (DEGs). RNA-Seq data changes depending on the tissue, the disease, and the time — unlike DNA which stays the same.

INTERVIEW TIP: RNA-Seq = measures which genes are active. DEG = differentially expressed gene. Cancer vs normal is the classic use.

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RNA-SEQ ANALYSIS

**What is DESeq2 and what does it find?**

DESeq2 is an R package used to find differentially expressed genes from RNA-Seq data. It compares gene counts between two groups (like treated vs untreated) and uses statistics to find which differences are real and not just due to random chance. A gene is called differentially expressed if its adjusted p-value is below 0.05 and its log2 fold change is above 1 (at least doubled) or below -1 (at least halved). DESeq2 also corrects for multiple testing automatically.

INTERVIEW TIP: DESeq2 finds which genes changed between two groups. $FDR < 0.05 + \log_2 FC > 1$ = significant. Remember both numbers.

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DATABASES & ANNOTATION

**What are NCBI, Ensembl, and UniProt? How are they different?**

These are three important biological databases. NCBI is the biggest — it stores DNA and RNA sequences (GenBank), variant information (dbSNP, ClinVar), research papers (PubMed), and more. Ensembl focuses on how genes are organised and annotated across different species — it is useful for looking up gene structures. UniProt stores information about proteins — what they do, where they work, and their structure. In bioinformatics, you use all three at different stages of your analysis.

INTERVIEW TIP: NCBI = everything (DNA, variants, papers). Ensembl = gene structure. UniProt = protein function. Three different uses.

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AI & MACHINE LEARNING

**How is machine learning used in bioinformatics?**

Machine learning is used to find patterns in large biological datasets that humans cannot find manually. Common uses include: predicting whether a variant is harmful or harmless, grouping cancer patients by their gene expression patterns, and predicting how a drug will bind to a protein. Tools like scikit-learn in Python are used with data tables where each row is a sample and each column is a biological measurement like a gene expression value.

INTERVIEW TIP: ML in bioinformatics = finding patterns in big data. Variant classification and cancer subtype prediction are the two most common examples.

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CLINICAL APPLICATIONS

**What is personalised medicine? How does genomics help?**

Personalised medicine means giving each patient a treatment that is chosen based on their own genetic information — not the same treatment for everyone. Genomics helps by showing which specific mutation is causing the disease. For example, two patients may both have lung cancer but one has an EGFR mutation and the other does not. The patient with the EGFR mutation will respond to a targeted drug. The patient without it will not. Without genomic testing, doctors might give the wrong treatment.

INTERVIEW TIP: Personalised medicine = right treatment for right mutation. EGFR mutation + targeted therapy = the classic example. Know it.

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CLINICAL APPLICATIONS

**What is the EGFR mutation and why does it matter in cancer?**

EGFR (Epidermal Growth Factor Receptor) is a protein that tells cells to grow. When the EGFR gene has a mutation, the protein gets stuck in the ON position and tells cells to grow without stopping — this causes cancer. In non-small cell lung cancer, the most common EGFR mutations are in exon 19 (deletion) and exon 21 (L858R). Patients with these mutations respond well to targeted drugs called EGFR inhibitors. A later mutation called T790M can make the cancer resistant to first-generation drugs.

INTERVIEW TIP: EGFR mutation = stuck ON signal = cancer. Exon 19 deletion + L858R = sensitive to targeted drugs. T790M = resistance.

What To Do With These Answers Now?

Reading the answers is step one. The next step is to practise saying them out loud. Do this the night before your interview — not five minutes before. Take each answer, say it in your own words, and make sure it feels natural. The goal is to sound like you understand it, not like you memorised it.

If you want to go deeper — to actually run NGS pipelines, analyse real data, and build the portfolio that makes interviewers call you back — that is what BioCode Mastery is for. It is an 8-week hands-on program where you learn by doing, not just reading.

+ BioCode Mastery Program	8-week hands-on: Python, R, Linux, NGS, AI — with real datasets and a project portfolio
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